

Evaluating the Clinical Utility of Multimodal Data for Early Sepsis Prediction: An Explainable Late-Fusion Approach

This document outlines the methodology of an empirical evaluation: whether multimodal late-fusion of tabular Electronic Health Records (EHR), Electrocardiograms (ECG), chest X-ray images (CXR), and radiology reports provides incremental value for early sepsis prediction in Intensive Care Unit (ICU) patients, beyond what EHR alone achieves. The pipeline targets prediction six hours before sepsis onset, uses frozen pre-trained foundation models for feature extraction, and combines modalities through an explainable late-fusion architecture designed to handle missing modalities and surface per-patient modality contributions. Aggregate results, ablations, and explainability analyses are reported in the full dissertation.

1 Cohort Selection & Dataset Creation

The study cohort is derived from the MIMIC-IV database. Sepsis onset is formally defined using the `mimiciv_derived.sepsis3` criteria (the first instance of a SOFA score ≥ 2 combined with suspected infection within the ICU).

Case Definitions

- **Positive Cases (Sepsis Positive):** Patients who developed sepsis at least 24 hours after hospital admission. The prediction target (Anchor Time) is strictly set to 6 hours before the official clinical onset to ensure proactive prediction.
- **Negative Cases (Control Group):** Patients with no history of sepsis and an adequate length of stay. To create a realistic prediction window, the median time from ICU admission to sepsis onset was calculated for the positive cohort, and this interval was applied to the negative patients to establish their pseudo-Anchor Time.

All multimodal data (vitals, labs, images, waveforms) used for inference strictly precede the Anchor Time to prevent data leakage.

Admission & Study Deduplication

To ensure strict patient-level independence (one admission per patient), the following prioritization logic was applied:

1. Admissions containing auxiliary modalities (CXR or ECG) were prioritized to maximize multimodal data availability.
2. If multiple admissions had the same modality availability, the chronologically oldest admission was selected.

For multiple studies within an admission:

- Only the latest valid study preceding the Anchor Time was retained.
- For CXR studies containing multiple images, priority was given to the capture position (PA > AP > Lateral), followed by the highest overall pixel resolution ($Rows \times Cols$).

Modality	Sepsis Positive	Sepsis Negative	Total Available
EHR (Tabular)	8,436	7,077	15,513
ECG	3,906	3,240	7,146
CXR Image & Report	1,511	701	2,212

Cohort Statistics

The final cohort comprises **15,513 unique patients**.

Note: The higher prevalence of sepsis in the CXR subgroup reflects realistic clinical triaging, where diagnostic imaging is preferentially ordered for patients exhibiting higher visual acuity and illness severity. Data splitting (Train/Val/Test) relies on strict stratified splitting to maintain these proportional distributions across both outcomes and modality combinations.

2 Pre-trained Foundation Models

Due to the limited volume of multimodal data available for training from scratch, high-dimensional feature extraction is handled by frozen pre-trained Foundation Models.

- **EHR (Tabular):** StanfordShahLab/motor-t-base
- **ECG:** wanglab/ecg-fm
- **CXR Image:** torchxrayvision/densenet121-res224-all
- **CXR Report (Text):** emilyalsentzer/Bio_ClinicalBERT

Note: These foundation models are used offline to generate fixed embeddings for each modality. The training code loads these pre-computed tensors; the foundation models themselves are not part of the end-to-end training pipeline.

3 Neural Network Architecture

The network is a dynamic late-fusion design summarized in Figure 1. The mathematical flow is described in the subsections below.

A. Input Preparation

Each foundation model produces a raw embedding $z_{i,raw}$ that is L2-normalized at extraction time:

$$z_{i,norm} = \frac{z_{i,raw}}{\|z_{i,raw}\|_2}$$

The fusion network receives these normalized vectors together with binary availability masks $M_i \in \{0, 1\}$ indicating per-patient modality presence.

B. Dimensional Projection

To align the disparate modalities, the normalized embeddings are linearly projected into a shared dimensional space:

$$z_{i,proj} = \text{Linear}_i(z_{i,norm})$$

C. Unimodal Prediction

To enforce unimodal accountability and encourage strong standalone features within the shared space, the projected embeddings are passed through independent Multi-Layer Perceptrons (MLP_i) to generate standalone sepsis probabilities:

$$\begin{aligned} \text{logit}_i &= MLP_i(z_{i,proj}) \\ p_i &= \sigma(\text{logit}_i) \end{aligned}$$

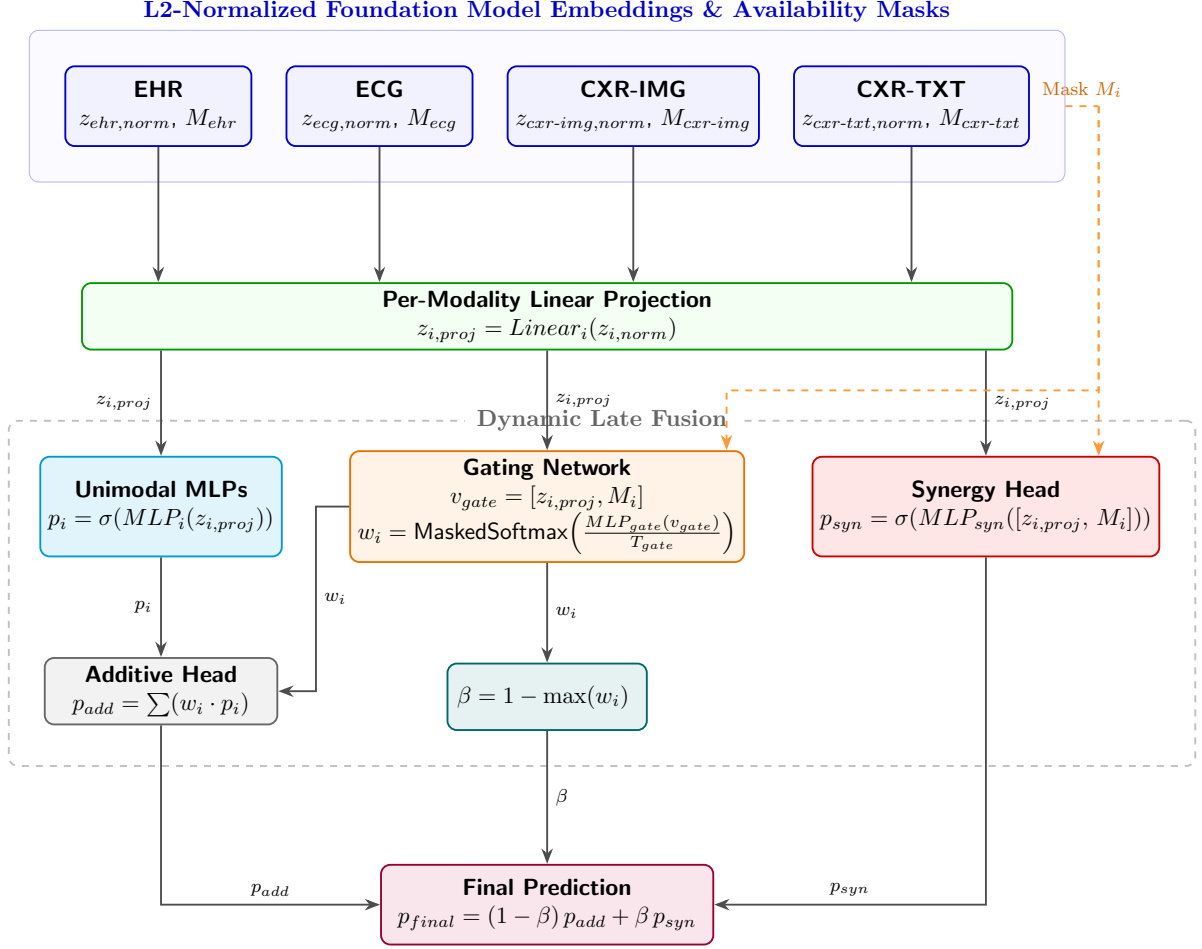


Figure 1: Architecture of the dynamic late-fusion sepsis-prediction model.

D. Gating Network

A specialized Gating Network determines the dynamic importance of each modality per patient. It evaluates the concatenated projected vectors alongside a binary availability mask (M):

$$v_{gate} = [z_{ehr,proj}, z_{ecg,proj}, z_{cxr-img,proj}, z_{cxr-txt,proj}, M]$$

$$logits_{gate} = MLP_{gate}(v_{gate})$$

Before determining the final weights, the raw logits are divided by a learnable gating temperature parameter (T_{gate}). This controls the sharpness of the modality attention distribution. A Masked Softmax function then zeroes out the weights of missing modalities ($M_i = 0$), ensuring the active weights sum to 1:

$$w_i = \frac{M_i \cdot \exp(logits_{gate,i}/T_{gate})}{\sum_j (M_j \cdot \exp(logits_{gate,j}/T_{gate}))}$$

E. Late Fusion (Additive & Synergy)

- **Additive Head:** Computes the weighted average of the independent unimodal probabilities.

$$p_{add} = \sum(w_i \cdot p_i)$$

- **Synergy Head:** Captures complex, systemic, non-linear interactions across modalities by processing the concatenated projected vectors alongside the binary availability mask (M).

$$p_{syn} = \sigma(MLP_{syn}([z_{ehr,proj}, z_{ecg,proj}, z_{cxr-img,proj}, z_{cxr-txt,proj}, M]))$$

F. Final Prediction

The final probability interpolates between the additive and synergistic heads using a dynamic mixing coefficient β , defined as $\beta = 1 - \max(w_i)$. This causes the model to rely on Synergy when signals are diffuse, and on Additive logic when a single modality dominates:

$$p_{final} = (1 - \beta) p_{add} + \beta p_{syn}$$

4 Training Objective & Loss

The network is optimized using a composite objective function that balances the final prediction error with a structural constraint enforcing independent unimodal accuracy:

$$L_{total} = BCE(p_{final}, y) + \lambda \sum_{i: M_i=1} BCE(p_i, y)$$

The auxiliary loss is computed by **summing** the unimodal BCE terms over only the active (present) modalities for each patient. Because EHR is available for all patients while the auxiliary modalities (ECG, CXR image, and CXR report) are present for only a minority of cases, this formulation naturally up-weights the auxiliary gradient for patients who have richer multimodal data — precisely the patients on which the gating network and synergy head benefit most from strong unimodal supervision.

The weighting λ is treated as a tunable hyperparameter (search range 0.05 to 0.6) and optimized empirically via Optuna on the validation set, rather than fixed to a static default.

5 Post-Hoc Calibration

To support interpretation of the model’s outputs as triaging confidences, Temperature Scaling is applied to the final fused logit. A scalar temperature parameter (T_{final}) is fitted on validation logits and used at inference time:

$$p_{final,cal} = \sigma\left(\frac{logit_{final}}{T_{final}}\right)$$

6 Explainability

Interpretability is integrated at both the architectural (macro) and feature (micro) levels.

Macro-Level: Modality Contribution

The gating network’s attention weights (w_i) provide a per-patient readout of the modality reliance the trained model has learned. Together with the synergy coefficient β , these weights can be analyzed at cohort scale to characterize how the architecture allocates predictive weight across the available modalities and how that allocation interacts with synergy versus additive logic.

Micro-Level: EHR Token Attribution

For per-patient transparency on the EHR branch, the repository implements Captum’s `LayerGradientXActivation` on the MOTOR transformer’s first-layer normalization output. The resulting attributions trace each predicted sepsis risk back to specific clinical tokens — lab results, vital sign categories, medications, and procedures — in the patient’s pre-anchor timeline.